ROBOT CAR PRINTED CIRCUIT BOARD DESIGN

STM32F103 MICROCONTROLLER AND TB6612FNG MOTOR DRIVER

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AKOKA, NIGERIA

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1.0 Background Study

The STM32F103 is a versatile and widely-used 32-bit microcontroller for embedded development, offering a balance of performance, features, and cost-effectiveness. It offers high performance and low power consumption, making it suitable for a wide range of embedded applications. The STM32F103 series features a rich set of peripherals including GPIO (General Purpose Input/Output) pins, UART (Universal Asynchronous Receiver/Transmitter), SPI (Serial Peripheral Interface), I2C (Inter-Integrated Circuit), ADC (Analog-to-Digital Converter), timers, PWM (Pulse Width Modulation), and more. The STM32F103 microcontroller can be programmed in STM32CubeIDE and C/C++ using libraries provided by STMicroelectronics or third-party libraries.
Due to its performance, peripherals, and low-power features, the STM32F103 microcontroller is commonly used in a wide range of embedded applications including industrial control systems, consumer electronics, smart appliances, medical devices, automotive systems, and more.

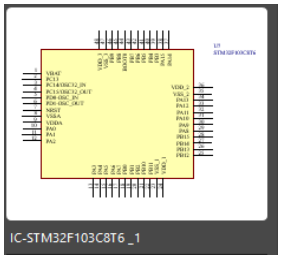
The TB6612FNG is a dual motor driver IC (Integrated Circuit) that is commonly used in robotics and other projects requiring motor control. It can control two DC motors. This dual functionality makes it versatile for various applications, including wheeled robots, robotic arms, and more. It operates at a wide range of voltages, typically from 2.7V to 5.5V. The IC incorporates current limit functionality, which protects the motors from damage due to overcurrent conditions. This feature ensures the longevity of both the motor and the driver IC itself. The TB6612FNG accepts PWM inputs for controlling the speed and direction of the motors it drives. It has a straightforward control interface, typically requiring only a few control pins to operate which makes it easy to integrate into various projects and microcontroller platforms. The TB6612FNG is available in compact packages, such as SSOP and QFN, which helps save space on PCBs (Printed Circuit Boards) and makes it suitable for applications with size constraints. Compared to more advanced motor drivers, the TB6612FNG provides a cost-effective solution for basic motor control needs without compromising on essential features.

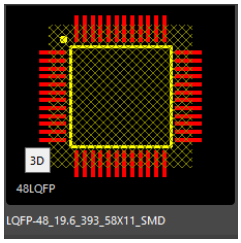
2.0 Aim and Objectives

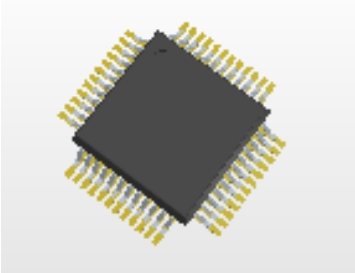
To design a Printed circuit Board for a robot car using STM32F103 Microcontroller and TB6612FNG motor driver using Altium designer

3.0 Apparatus/Component Used

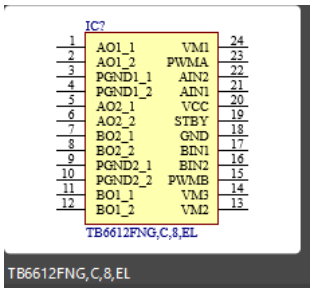
1. STM32F103C8T6 Microcontroller

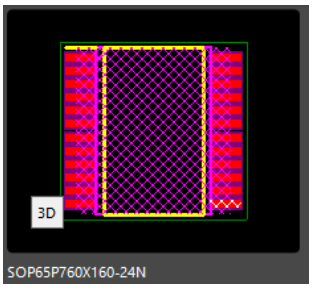


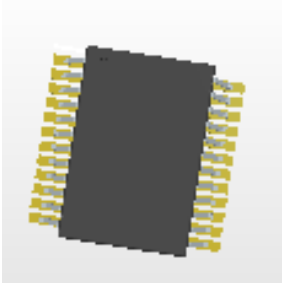




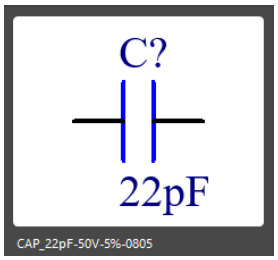
2. TB6612FNG Motor Driver IC

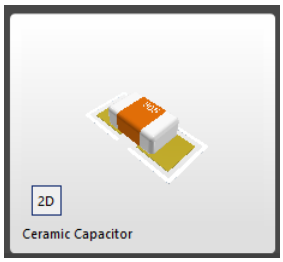


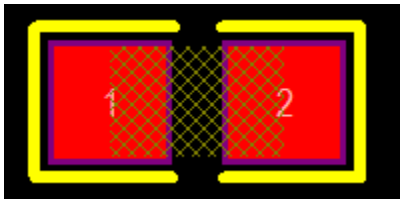




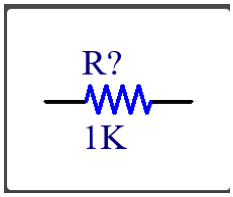
3. Capacitors (0.1uF,1uF,22pF)

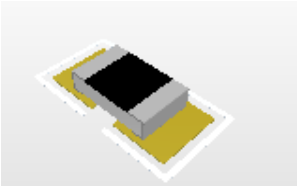
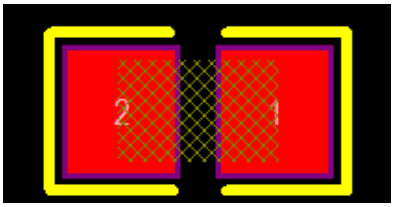




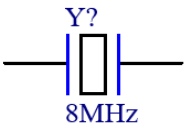


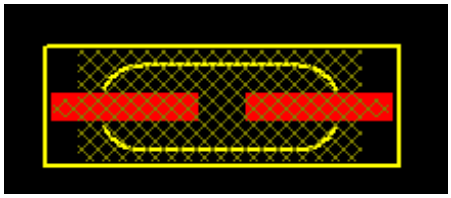
4. Resistors (10k ohms,1k ohms)



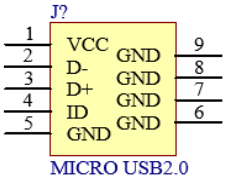



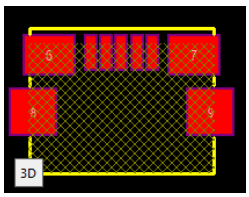
5. Crystal Oscillators (32.768KHz,8Mhz)



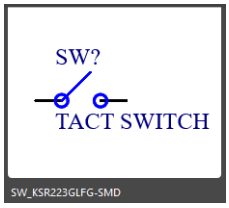


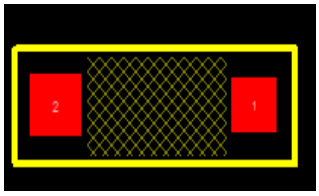
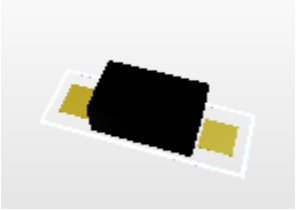
6. Connectors (2x3, 20x1, micro USB)



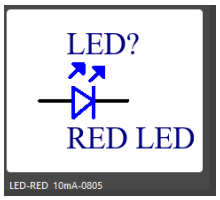


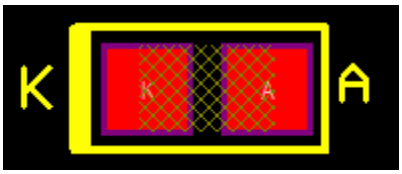
7. Switch

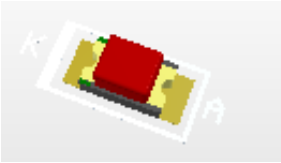


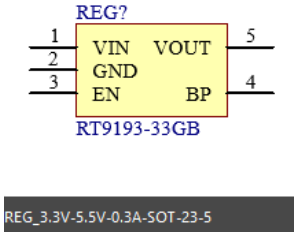
8. LEDs (Red)

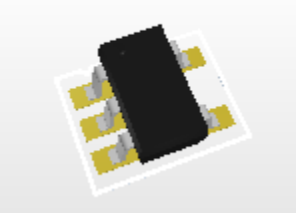
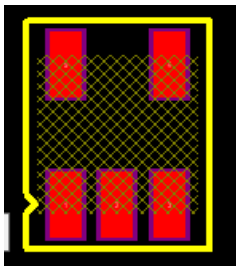






9. Regulator (5v-3.3v)



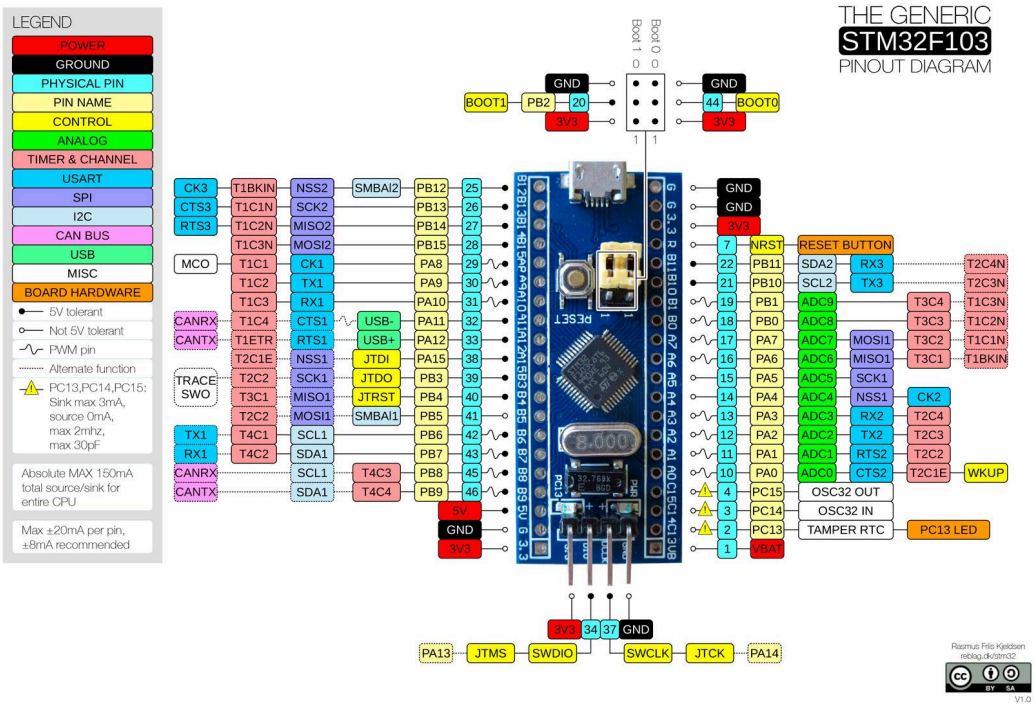



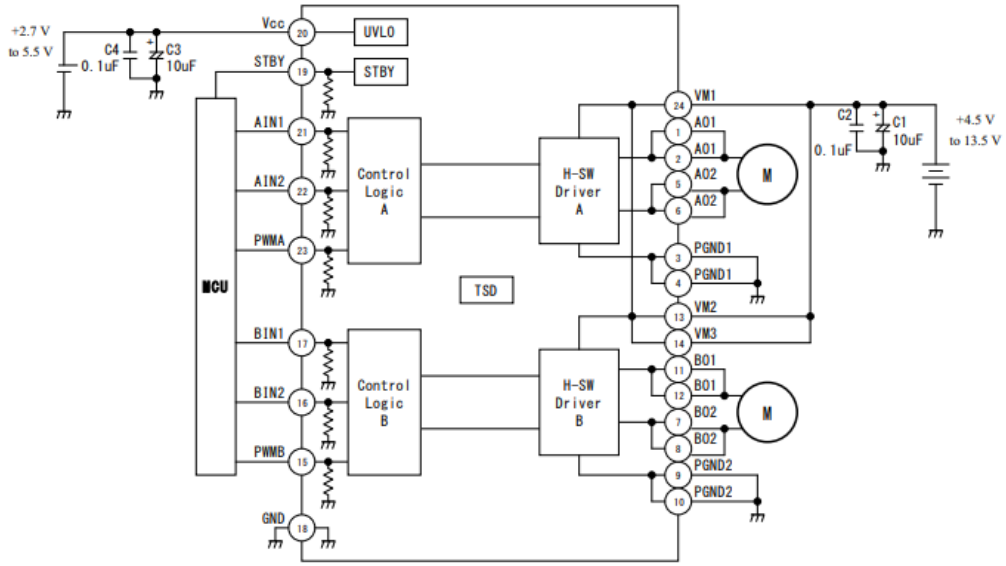
4.0 METHODOLOGY/ PROCEDURE

1. Create the components (symbol and footprint) in a library or download a library with the components needed and compile the library.
2. Create a new project, add new schematic and PCB document
3. On the schematic page, place the components and connect according to the circuit diagram for STM32F103 and TB6612FNG
4. Annotate the components on the schematic page
5. Import the schematic to the PCB document
6. Resize the PCB to the appropriate size
7. Place the components properly on the PCB board
8. Make the connection using wires(track) and vias on the PCB board
9. For the ground connection,do a polygon pour
10. Prepare the job output file to be sent company for the PCB printing (and assembly)

5.0 Diagrams

5.1 Circuit Diagram/ Pin out

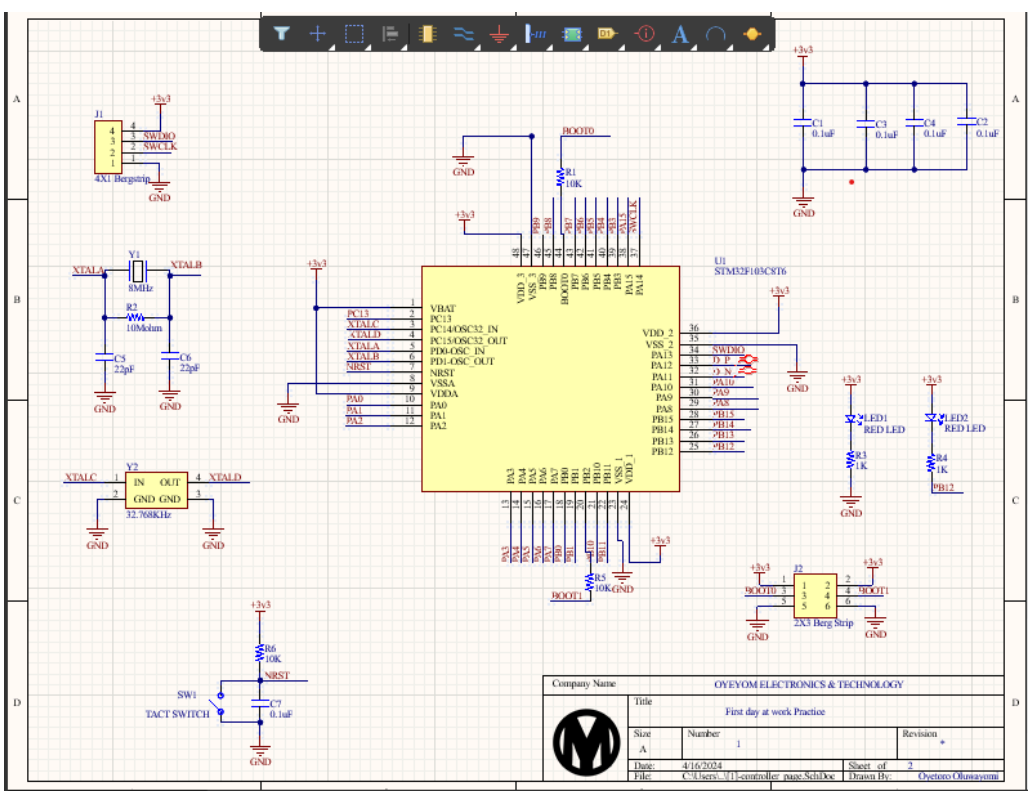


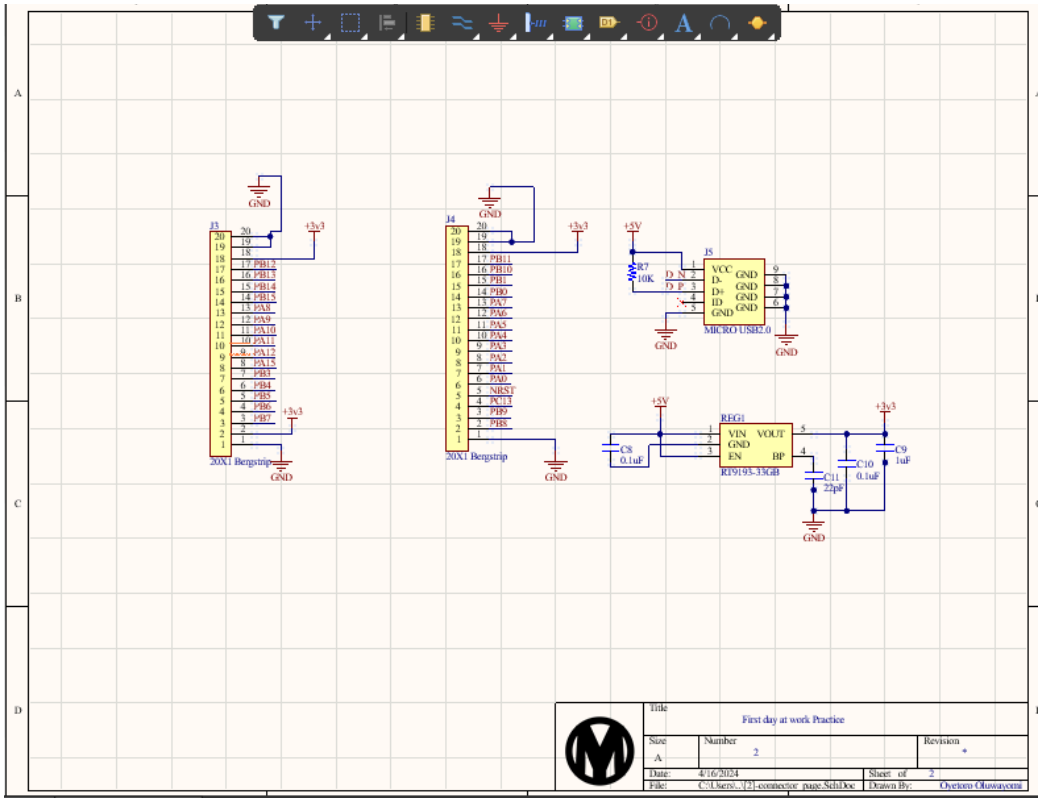


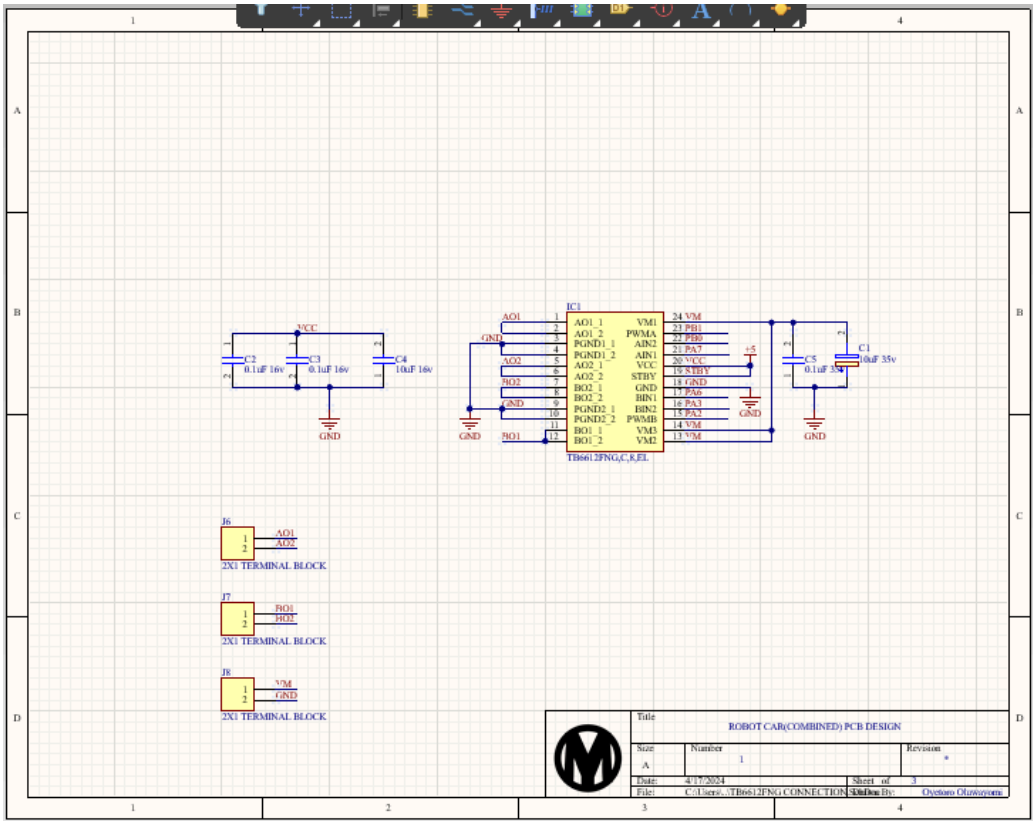
5.2 PCB board size

Considering the size of the chassis 70.237mm by 25.4mm PCB was used.

5.3 Schematic Diagram

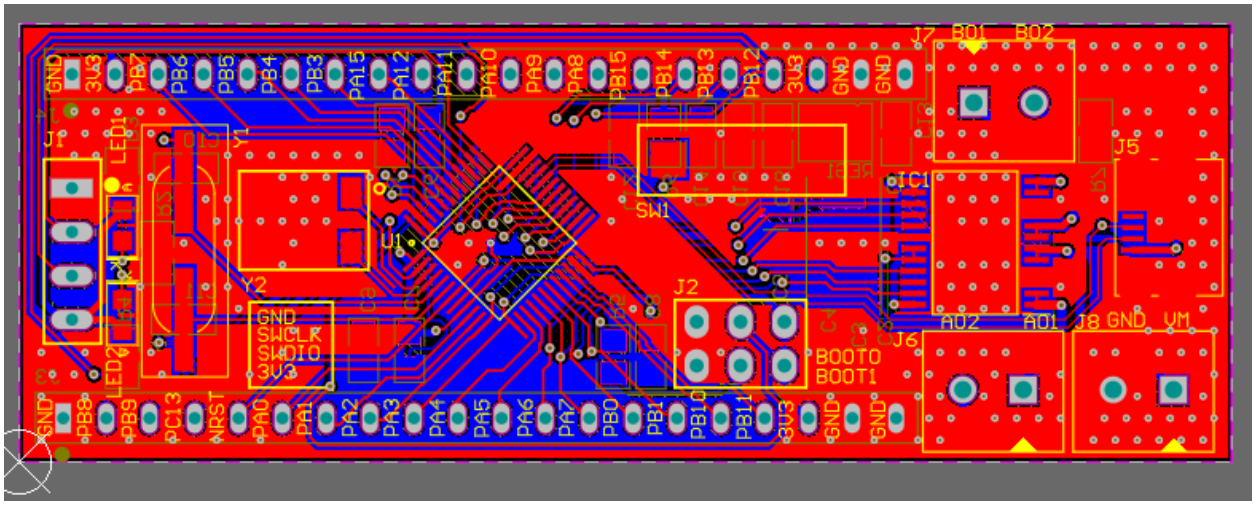



J3
20X1 Bengstrip
J4
20X1 Bengstrip
J5
MICRO USB2.0
R7
10K
REG1
RT9193-33GB
C8
0.1uF
C9
1uF
C10
0.1uF
C11
22pF
GND
+3v3
+5V
Title
First day at work Practice
Size
A
Number
2
Revision
*
Date: 4/16/2024
Sheet of 2
File: C:\Users\...\[2]-connector_page.SchDoc
Drawn By: Oyetoro Oluwayomi

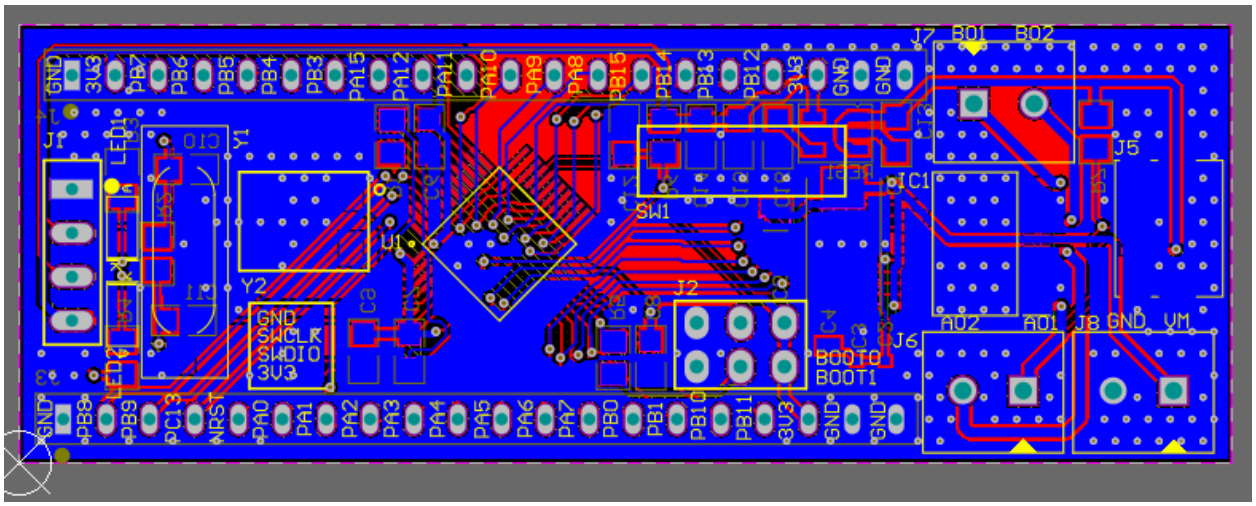

IC1
TB6612FNG,C,K,EL
C2
0.1uF 16v
C3
0.1uF 16v
C4
10uF 16v
C5
0.1uF 35v
C1
10uF 35v
VCC
GND
+5
J6
2X1 TERMINAL BLOCK
J7
2X1 TERMINAL BLOCK
J8
2X1 TERMINAL BLOCK
Title
ROBOT CAR(COMBINED) PCB DESIGN
Size
A
Number
1
Revision
*
Date: 4/17/2024
Sheet of 3
File: C:\Users\...\TB6612FNG CONNECTION.SchDoc
Drawn By: Oyetoro Oluwayomi

5.4 PCB Board

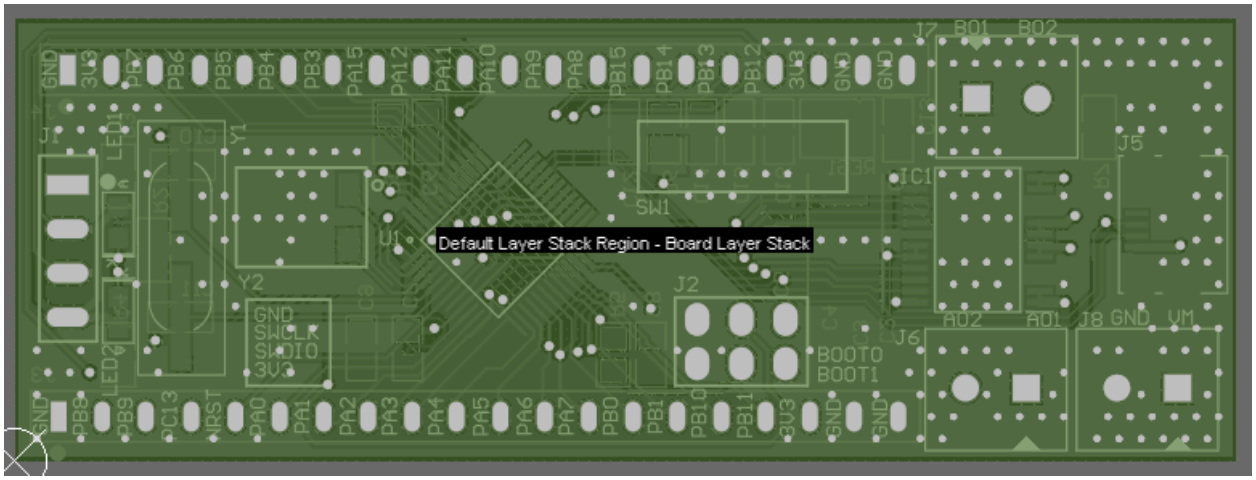
Top layer



Bottom layer

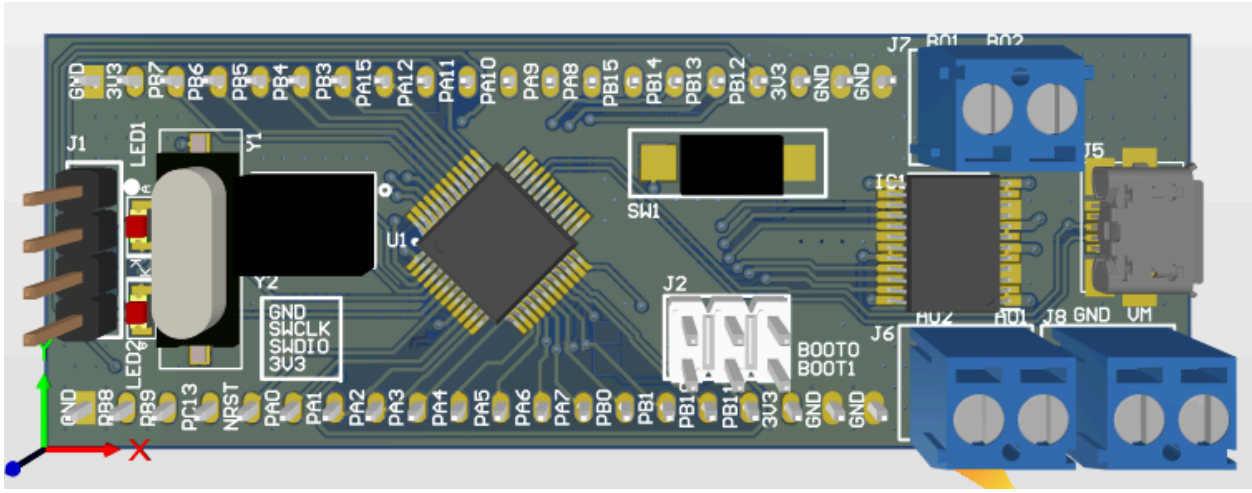


Layer stack

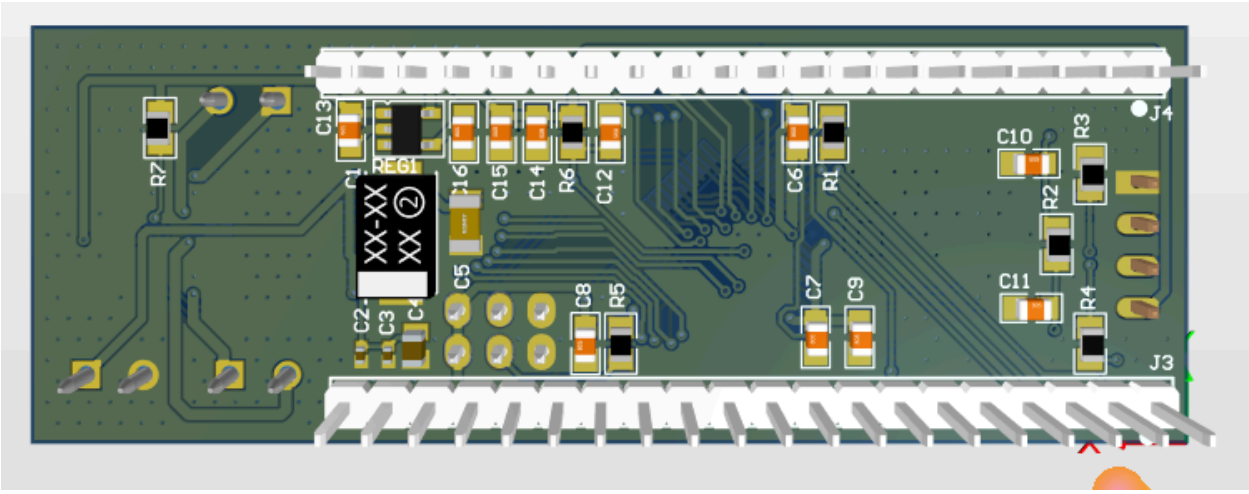


6.0 3D PCB BOARD

3D Top layer



3D Bottom Layer



7.0 CONCLUSION

The Printed Circuit board(combined) consisting of the STM32F103 microcontroller and TB6612FNG motor driver for the robot car was successfully designed and ready for printing.